Data Science Projects: Beginner to Advanced

With Problem Statements, Solutions, and Dummy Datasets

Here is a structured project roadmap you can use to practice **Python, NumPy, Pandas, Matplotlib, Seaborn, Machine Learning, Deep Learning, and Data Visualization**.

LEVEL 1: BEGINNER PROJECTS

1. Student Marks Analysis

Concepts

- Python
- Pandas
- CSV files
- Mean, maximum, minimum
- Data visualization

Dummy Data

csv

```csv
Student_ID,Name,Maths,Science,English
1,Asha,85,78,90
2,Ravi,70,82,75
3,Priya,95,88,92
4,Kumar,60,65,70
5,Anu,88,90,85
```

Python Solution

Python Run

```python
import pandas as pd

df = pd.read_csv("students.csv")

print(df)

print("Average Marks:")
print(df[["Maths", "Science", "English"]].mean())

df["Total"] = df["Maths"] + df["Science"] + df["English"]

df["Average"] = df["Total"] / 3
```

```
print(df)
```

Practice Tasks

1. Find the top student.
2. Find the lowest student.
3. Calculate average marks.
4. Add a grade column.
5. Create a bar chart.

---

2. Sales Data Analysis

Dummy Data

csv

```
Order_ID,Product,Category,Quantity,Price
101,Laptop,Electronics,2,50000
102,Mouse,Accessories,5,500
103,Keyboard,Accessories,3,1500
104,Mobile,Electronics,2,25000
105,Headphones,Accessories,4,2000
```

Solution

Python Run

```
import pandas as pd

df = pd.read_csv("sales.csv")

df["Total_Sales"] = df["Quantity"] * df["Price"]

print(df)

print("Total Revenue:")
print(df["Total_Sales"].sum())

print("Category Sales:")
print(df.groupby("Category")["Total_Sales"].sum())
```

Visualization

Python Run

```python
import matplotlib.pyplot as plt

category_sales = df.groupby("Category")["Total_Sales"].sum()

category_sales.plot(kind="bar")

plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")

plt.show()
```

---

3. Employee Salary Analysis

Dummy Data

csv

```csv
Employee_ID,Name,Department,Salary,Experience
1,Raj,IT,50000,2
2,Priya,HR,45000,3
3,Arun,IT,65000,5
4,Divya,Finance,55000,4
5,Karthik,IT,70000,6
```

Solution

Python Run

```python
import pandas as pd

df = pd.read_csv("employees.csv")

print(df.describe())

print("Average Salary:")
print(df["Salary"].mean())

print("Department Wise Salary:")
print(df.groupby("Department")["Salary"].mean())
```

---

4. Customer Data Cleaning Project

Dummy Data

csv

```csv
Customer_ID,Name,Age,City,Salary
1,Ravi,25,Chennai,30000
2,Priya,,Madurai,40000
3,,30,Chennai,35000
4,Arun,28,,45000
5,Divya,25,Madurai,
```

Solution

Python Run

```python
import pandas as pd

df = pd.read_csv("customers.csv")

print(df.isnull().sum())

df["Age"] = df["Age"].fillna(df["Age"].mean())

df["Salary"] = df["Salary"].fillna(df["Salary"].mean())

df["Name"] = df["Name"].fillna("Unknown")

df["City"] = df["City"].fillna("Unknown")

print(df)
```

Concepts Learned

- Missing values
- `fillna()`
- `isnull()`
- Data cleaning

LEVEL 2: INTERMEDIATE PROJECTS

5. E-Commerce Sales Dashboard

Dummy Data

csv

```csv
Order_ID,Date,Product,Category,Quantity,Price,Customer_City
1,2026-01-01,Laptop,Electronics,1,50000,Chennai
```

```
2,2026-01-02,Mouse,Accessories,3,500,Madurai
3,2026-01-03,Mobile,Electronics,2,25000,Coimbatore
4,2026-01-04,Keyboard,Accessories,4,1500,Chennai
5,2026-01-05,Laptop,Electronics,1,50000,Madurai
```

Tasks

- Calculate total revenue.
- Find best-selling products.
- Find top city.
- Monthly sales analysis.
- Category-wise sales.
- Create charts.

Solution

Python Run

```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("ecommerce.csv")

df["Revenue"] = df["Quantity"] * df["Price"]

print("Total Revenue")
print(df["Revenue"].sum())

print("Best Selling Products")
print(df.groupby("Product")["Quantity"].sum())

print("Sales by City")
print(df.groupby("Customer_City")["Revenue"].sum())

df.groupby("Category")["Revenue"].sum().plot(
    kind="pie",
    autopct="%1.1f%%"
)

plt.title("Category Revenue")

plt.show()
```

---

6. Netflix Movies Data Analysis

Dummy Data

csv

```csv
Movie_ID,Title,Genre,Year,Rating,Duration
1,Movie A,Action,2022,8.5,120
2,Movie B,Comedy,2023,7.2,100
3,Movie C,Drama,2021,9.0,150
4,Movie D,Action,2023,8.0,130
5,Movie E,Comedy,2022,6.8,95
```

Tasks

1. Find the highest-rated movie.
2. Find the average rating.
3. Count movies by genre.
4. Find movies released after 2022.
5. Create genre visualization.

Solution

Python Run

```python
import pandas as pd

df = pd.read_csv("movies.csv")

print(df.loc[df["Rating"].idxmax()])

print("Average Rating:")
print(df["Rating"].mean())

print(df["Genre"].value_counts())

new_movies = df[df["Year"] >= 2022]

print(new_movies)
```

---

7. Bank Customer Churn Analysis

Dummy Data

csv

```csv
Customer_ID,Age,Salary,Account_Balance,Years,Churn
1,25,30000,50000,2,0
2,45,70000,150000,10,0
3,30,40000,20000,1,1
```

```
4,50,90000,300000,15,0
5,28,35000,10000,1,1
```

Machine Learning Problem

Predict whether a customer will leave the bank.

Solution

Python Run

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

df = pd.read_csv("bank.csv")

X = df[[
    "Age",
    "Salary",
    "Account_Balance",
    "Years"
]]

y = df["Churn"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = RandomForestClassifier()

model.fit(X_train, y_train)

prediction = model.predict(X_test)

print(prediction)
```

---

8. House Price Prediction

Dummy Data

csv

```
Area,Bedrooms,Bathrooms,Age,Price
1000,2,1,10,3000000
1500,3,2,5,5000000
2000,4,3,2,8000000
1200,2,2,8,4000000
1800,3,2,4,6500000
```

Model

Linear Regression

Solution

Python Run

```python
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

df = pd.read_csv("house.csv")

X = df[[
    "Area",
    "Bedrooms",
    "Bathrooms",
    "Age"
]]

y = df["Price"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

prediction = model.predict(X_test)

print(prediction)
```

Predict a New House

Python Run

```
new_house = [[1600, 3, 2, 5]]

price = model.predict(new_house)

print(price)
```

LEVEL 3: MACHINE LEARNING PROJECTS

9. Student Performance Prediction

Features

- Study hours
- Attendance
- Previous marks
- Sleep hours

Dummy Data

csv

```
Study_Hours,Attendance,Previous_Marks,Sleep_Hours,Final_Marks
2,70,60,6,65
4,80,70,7,75
6,90,80,8,88
3,75,65,6,70
8,95,90,7,96
```

Solution

Python Run

```
import pandas as pd

from sklearn.linear_model import LinearRegression

df = pd.read_csv("student_performance.csv")

X = df[[
    "Study_Hours",
    "Attendance",
    "Previous_Marks",
    "Sleep_Hours"
]]

y = df["Final_Marks"]
```

```python
model = LinearRegression()

model.fit(X, y)

new_student = [[5, 85, 75, 7]]

prediction = model.predict(new_student)

print("Predicted Marks:", prediction)
```

---

10. Spam Email Detection

Dummy Data

csv

```csv
Message,Spam
"Win a free iPhone now",1
"Meeting scheduled tomorrow",0
"Congratulations you won money",1
"Please submit your assignment",0
"Claim your prize now",1
```

Concepts

- NLP
- Text preprocessing
- CountVectorizer
- Classification

Solution

Python Run

```python
import pandas as pd

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

df = pd.read_csv("spam.csv")

vectorizer = CountVectorizer()

X = vectorizer.fit_transform(df["Message"])

y = df["Spam"]
```

```
model = MultinomialNB()

model.fit(X, y)

new_message = vectorizer.transform([
    "Congratulations you won a prize"
])

prediction = model.predict(new_message)

print(prediction)
```

---

11. Loan Approval Prediction

Dummy Data

csv

```
Age,Income,Credit_Score,Loan_Amount,Approved
25,30000,650,200000,0
35,60000,750,500000,1
45,80000,800,700000,1
22,25000,600,150000,0
40,70000,720,400000,1
```

Algorithms to Try

- Logistic Regression
- Decision Tree
- Random Forest
- KNN

Solution

Python Run

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split

X = df[
    [
        "Age",
        "Income",
        "Credit_Score",
        "Loan_Amount"
    ]
]
```

```
y = df["Approved"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2
)

model = RandomForestClassifier()

model.fit(X_train, y_train)

print(model.score(X_test, y_test))
```

LEVEL 4: ADVANCED DATA SCIENCE PROJECTS

12. Customer Segmentation

Goal

Divide customers into different groups based on:

- Income
- Spending score
- Age
- Purchase frequency

Dummy Data

csv

```
Customer_ID,Age,Income,Spending_Score
1,25,30000,60
2,35,70000,80
3,45,100000,30
4,22,25000,90
5,50,90000,40
6,30,50000,70
```

Algorithm

K-Means Clustering

Solution

Python Run

```python
import pandas as pd

from sklearn.cluster import KMeans

df = pd.read_csv("customers.csv")

X = df[[
    "Income",
    "Spending_Score"
]]

model = KMeans(
    n_clusters=3,
    random_state=42
)

df["Cluster"] = model.fit_predict(X)

print(df)
```

---

13. Sales Forecasting Project

Dummy Data

csv

```csv
Date,Sales
2026-01-01,10000
2026-01-02,12000
2026-01-03,15000
2026-01-04,13000
2026-01-05,17000
2026-01-06,18000
```

Concepts

- Time series
- Trend analysis
- Forecasting

Basic Solution

Python Run

```python
import pandas as pd

df = pd.read_csv("sales.csv")

df["Date"] = pd.to_datetime(df["Date"])

df = df.sort_values("Date")

df["Previous_Sales"] = df["Sales"].shift(1)

print(df)
```

Advanced Algorithms

- ARIMA
- SARIMA
- Prophet
- LSTM

---

14. Credit Card Fraud Detection

Problem

Predict whether a transaction is:

```
0 = Normal
1 = Fraud
```

Dummy Data

csv

```
Amount,Hour,Location_Risk,Transaction_Count,Fraud
500,10,0,2,0
100000,2,1,10,1
300,14,0,1,0
50000,3,1,8,1
1000,12,0,3,0
```

Algorithms

- Logistic Regression
- Random Forest
- XGBoost
- Isolation Forest

Important Concepts

- Imbalanced data
- Precision
- Recall
- F1 Score
- Confusion Matrix

---

15. Recommendation System

Example: Movie Recommendation System

Dummy Data

```csv
User_ID,Movie_ID,Rating
1,101,5
1,102,4
1,103,2
2,101,4
2,102,5
3,103,5
3,104,4
```

Concepts

- Collaborative filtering
- User similarity
- Item similarity
- Recommendation algorithms

---

16. Sentiment Analysis Project

Dummy Data

```csv
Review,Sentiment
"This product is amazing",Positive
"I hate this product",Negative
"Very good quality",Positive
```

"Not worth the money" Negative

Solution

Python Run

```python
import pandas as pd

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

df = pd.read_csv("reviews.csv")

vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(df["Review"])

y = df["Sentiment"]

model = LogisticRegression()

model.fit(X, y)

test = vectorizer.transform([
    "This product is excellent"
])

print(model.predict(test))
```

LEVEL 5: DEEP LEARNING PROJECTS

17. Handwritten Digit Recognition

Dataset

Use the built-in MNIST dataset.

Concepts

- Neural Networks
- TensorFlow
- Keras
- CNN

Basic Solution

Python Run

```
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten

(X_train, y_train), (X_test, y_test) = mnist.load_data()

X_train = X_train / 255.0
X_test = X_test / 255.0

model = Sequential([
    Flatten(input_shape=(28, 28)),
    Dense(128, activation="relu"),
    Dense(10, activation="softmax")
])

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

model.fit(
    X_train,
    y_train,
    epochs=5
)

model.evaluate(
    X_test,
    y_test
)
```

---

18. Image Classification

Example Categories

```
Cat
Dog
Car
Flower
```

Project Flow

```
Images
   ↓
Data Preprocessing
   ↓
```

```
Train/Test Split
   ↓
CNN Model
   ↓
Training
   ↓
Prediction
```

Technologies

- Python
- TensorFlow
- Keras
- OpenCV
- CNN

---

19. Fake News Detection

Dataset

```csv
News,Label
"Government announces new policy",REAL
"You will become rich overnight",FAKE
"New education scheme launched",REAL
"Secret method to earn millions",FAKE
```

Steps

```
1. Load Dataset
2. Clean Text
3. Convert Text to Numbers
4. Train Model
5. Test Accuracy
6. Predict New News
```

Algorithms

- Naive Bayes
- Logistic Regression
- Random Forest
- LSTM
- BERT

---

20. Complete Data Science Capstone Project

Smart E-Commerce Analytics and Prediction System

Features

1. Sales Analysis

```
Total Sales
Monthly Sales
Product Performance
Category Performance
```

2. Customer Analysis

```
Customer Segmentation
Top Customers
Purchase Frequency
Customer Lifetime Value
```

3. Prediction

```
Sales Prediction
Customer Churn Prediction
Product Demand Prediction
```

4. Dashboard

Use:

```
Streamlit
Power BI
Tableau
Plotly
Matplotlib
```

Recommended Learning Path

```
PYTHON
   ↓
NUMPY
   ↓
PANDAS
   ↓
```

```
DATA CLEANING
   ↓
DATA VISUALIZATION
   ↓
STATISTICS
   ↓
SQL
   ↓
MACHINE LEARNING
   ↓
MODEL EVALUATION
   ↓
FEATURE ENGINEERING
   ↓
DEEP LEARNING
   ↓
NLP
   ↓
COMPUTER VISION
   ↓
TIME SERIES
   ↓
DEPLOYMENT
```

Suggested 20-Project Challenge

Beginner

1. Student Marks Analysis
2. Employee Salary Analysis
3. Sales Data Analysis
4. Customer Data Cleaning
5. Weather Data Analysis

Intermediate

6. E-Commerce Dashboard
7. Netflix Data Analysis
8. Hospital Data Analysis
9. HR Analytics
10. Stock Data Visualization

Machine Learning

11. House Price Prediction
12. Student Performance Prediction
13. Loan Approval Prediction
14. Customer Churn Prediction
15. Spam Detection

Advanced

16. Customer Segmentation
17. Sales Forecasting
18. Fraud Detection
19. Recommendation System
20. Sentiment Analysis + Deep Learning

A good next step is to build these projects in order, with each project following this structure:

```
Project
│
├── dataset.csv
├── analysis.ipynb
├── data_cleaning.py
├── visualization.py
├── model.py
├── README.md
└── requirements.txt
```

For serious practice, use larger dummy datasets rather than only 5 rows—for example **100–1,000 rows** for beginner/intermediate projects and **5,000+ rows** for ML projects—so you can practice cleaning, visualization, training, and evaluation realistically.